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Review

Red and processed meat consumption and cancer outcomes: Umbrella review

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Highlights

- Umbrella review of various cancer risks related to red and processed meat intake.
- 100 g/d increment of red meat could increase by 11–51% risk of multiple cancer.
- 50 g/d increment of processed meat could increase by 8–72% risk of multiple cancer.
- Red and processed meat consumption seems to be not related to any benefit of cancer.

Abstract

The purpose of this umbrella review was to evaluate the quality of evidence, validity and biases of the associations between red and processed meat consumption and multiple cancer outcomes according to existing systematic reviews and meta-analyses. The umbrella review identified 72 meta-analyses with 20 unique outcomes for red meat and 19 unique outcomes for processed meat. Red meat consumption was associated with increased risk of overall cancer mortality, non-Hodgkin lymphoma (NHL), bladder, breast, colorectal, endometrial, esophageal, gastric, lung and nasopharyngeal cancer. Processed meat consumption might increase the risk of overall cancer mortality, NHL, bladder, breast, colorectal, esophageal, gastric, nasopharyngeal, oral cavity and oropharynx and prostate cancer. Dose-response analyses revealed that 100g/d increment of red meat and 50g/d increment of processed meat consumption were associated with 11%-51% and 8%-72% higher risk of multiple cancer outcomes, respectively, and seemed to be not correlated with any benefit.

Introduction

Meat is a vital source of essential nutrients in people's diets in many regions around the world (Speedy, 2003; X. Wang et al., 2016), including protein (Pereira and Vicente, 2013, Wyness, 2016), fat (Higgs, 2000), fatty acid (Krauss et al., 2000), vitamins (Williams, 2007) and minerals such as iron, zinc, and magnesium (Wyness, 2016). On the other hand, meat is also rich in some chemical substances such as heterocyclic amines (HCAs), polycyclic aromatic hydrocarbons (PAHs), nitrate and *N*-nitroso compounds (NOCs), which have been found to be associated with increased risk of multiple cancer (Mirvish, 1995, Moorthy et al., 2015, Sugimura et al., 2004). Given the continuous economic development, the need for meat is still on the rise both in developed and developing countries (Speedy, 2003, Walker et al., 2005). According to the International Agency for Research on Cancer (IARC) classification, red meat consumption was categorised as a probable human carcinogen based on the evidence for colorectal, pancreatic, and prostate cancer (Group 2A), while processed meat consumption was classified as carcinogenic to humans based on the evidence of colorectal cancer (Group 1) (Bouvard et al., 2015). As a result, the World Cancer Research Fund International (WCRF) recommended limiting the consumption of red meat to <500g per week, and consuming as little processed meat as possible (WCRF, 2018).

Recently, the associations between red and processed meat consumption and various cancer outcomes have been assessed in a large number of studies, including lots of systematic reviews and *meta*-analyses with diverse quality (Bouvard et al., 2015, Han et al., 2019, Händel et al., 2020). However, a comprehensive overview of multiple cancer outcomes is still deficient in literature, and few existing studies have assessed the quality of evidence for the associations of red and processed meat consumption with various

cancer outcomes (Crippa et al., 2018, Farvid et al., 2018, Han et al., 2019). Besides, the dose–response relationship between red and processed meat consumption and risk of cancer is still inconsistent in studies with various types of exposure (Schwingshackl et al., 2018, Wallin et al., 2011, Zhao et al., 2017). Therefore, we were aimed at performing a systematic evaluation of the associations between red and processed meat and multiple cancer outcomes that were reported in systematic reviews and *meta*-analyses.

To provide a general evaluation of the quality of evidence, possible biases and validity of the associations between red and processed meat consumption and diverse cancer outcomes, we performed this umbrella review of the evidence according to existing systematic reviews and *meta*-analyses.

Section snippets

Umbrella review methods

We systematically searched, organised and evaluated existing data from numerous systematic reviews and *meta*-analyses on multiple cancer outcomes associated with red and processed meat exposure (Aromataris et al., 2015, Papatheodorou, 2019).

Consumption of red and processed meat could be measured by servings or weight per day in *meta*-analyses. So, systematic reviews without *meta*-analysis were excluded in our study (Poole, Kennedy, Roderick, Fallowfield, Hayes, & Parkes, 2017)....

Literature search

We searched...

Characteristics of *meta*-analyses

Fig. 1 presents the flowchart of the literature search and selection process. We identified 1448 articles and yielded 72 *meta*-analyses on the basis of the inclusion and exclusion criteria. We extracted 20 unique cancer outcomes related to red meat consumption and 19 unique cancer outcomes related to processed meat consumption from eligible studies (Fig. 2). The associations of red meat consumption with cancer outcomes are presented in Table 1. Table 2 shows the associations between processed...

Principal findings and interpretation

In this umbrella review, a total of 72 *meta*-analyses with 20 unique outcomes related to red meat consumption and 19 unique outcomes related to processed meat consumption

were identified. Red meat consumption was associated with increased risk for bladder cancer, breast cancer, colorectal cancer, endometrial cancer, esophageal cancer, gastric cancer, lung cancer, nasopharyngeal carcinoma, NHL and overall cancer mortality. Processed meat consumption might increase the risk of bladder cancer,...

Conclusions

In summary, red and processed meat consumption was harmfully associated with a range of cancer outcomes and seemed to be not associated with any benefit in this umbrella review. Given the wide range of evidence for red and processed meat intake and various cancer risks, we recommend limiting the consumption of red meat to less than 100g per day and 500g per week, and consuming as little processed meat as possible. Further large-scale, high-quality prospective studies are required to confirm...

Individual contribution

YH and DHC collected, analyzed data and wrote the manuscript. LRL and QW designed the study, supervised the project, and revised the manuscript. BC, JL and QD assisted with detailed statistical analysis. ZYC and JBG helped with data extraction. All authors read and approved the final draft....

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper....

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